



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetsenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 7: Project Scope and Stakeholders

BSS 310 Engineering Management
2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- the basics about a marketing system. This include understanding and being able to apply knowledge from the definition, operations and characteristics of marketing systems to the semester project;
- the rules that govern customer service and care in business systems including measurement factors for service quality as affiliated to the marketing department.

Learning Outcomes for the Current Lecture

The expected learning outcomes at the end of this lecture include:

- Having a clear understanding of what a project scope and a scope statement are
- Understanding the significance of a project scope in a project
- Understanding the linkage between a project scope and stakeholders, including the stakeholders map, and
- Being able to understand, analyse, and create a business model canvas based on a clear understanding of objectives, scope, limitations, and delimitations
- Ability to adapt this knowledge to the semester project

BLOOM'S TAXONOMY & EXPECTED LEVELS of COGNITION

BLOOM'S TAXONOMY



Presentation Outline

- Definition of a project scope and a scope statement
- Significance of a project scope
- Project scope and stakeholders
- Stakeholders map: Who needs or influences what?
- Creating a Business Model Canvas

Definition of a Project Scope

- A project scope is a description or definition of the boundaries of the project. It defines the required work to complete a project.
- A scope is usually the "to-do list" that revolves around the objectives of a project.

Definition of a Project Scope

Watch and analyse a short YouTube video in class
about accepting a **project** without knowing the detailed
written **scope** of work.

<https://www.youtube.com/watch?v=wlnsbWtNUf0>

Definition of a Scope Statement

- A scope statement, is a written document that explicitly describes a project scope. It describes the main deliverable(s), assumptions and constraints.
- Scope statement provides a common understanding of the extent to which a project would be addressed among the stakeholders; it discusses what precise tasks and activities need to be addressed etc.

Scope Statement.....Contd.

- Scope statement states the main deliverables of a project
- Scope statement forms the basis of the contract between the project supplier and the end customer (internal or external)
- At the project closure the scope statement is used to determine whether the project has been acceptably completed



Scope Statement in the context of performance, time and operationability

<i>Non measurable</i>	<i>Measurable</i>
A new registration system	With the new system, students will be able to register for classes using a browser or a smartphone
The system will be fast	Response time will be under 1 second
The system will support the required volumes	75 students must be able to add an average of 3 classes each 15 minute period
Happy users	User satisfaction will be measured by a standardised set of questions and will measure 5 or higher on a 7 point scale
A new registration system	With the new system, students will be able to register for classes using a browser or a smartphone

<i>Property</i>	<i>Measure</i>
Size	MB/GB throughput
Ease of use	User acceptance test Help/support system
Reliability	Rate of failure occurrence Availability Mean time to failure Number of occurrences
Robustness	Time to restart after failure Number of events that cause failure
Scalability	Number of users

Scope Statement: A Generic Template

Project Scope Statement Template			
Project Name	Name of project		
Project Sponsor	Name of sponsor	Project Manager	Name of project manager
Date of Project Approval	Date of approval	Last Revision Date	Last updated date
Scope Description	List at a high level what is in the scope of the project as well as what is out of scope of the project.		
Project Deliverables	List the top level deliverables of the project		
Acceptance Criteria	List the project acceptance criteria.		

Scope Statement: A Generic Template..Contd.

Constraints	List any constraints that affect the project.
Assumptions	List any assumptions that the project is being based upon.

WE OFFER 3 KINDS OF SERVICES
GOOD - CHEAP - FAST

BUT YOU CAN ONLY PICK TWO

GOOD & CHEAP WON'T BE **FAST**

FAST & GOOD WON'T BE **CHEAP**

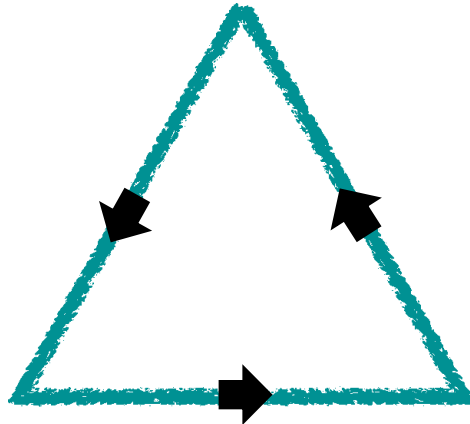
CHEAP & FAST WON'T BE **GOOD**

Significance of a Project Scope

The Project Triangle

Scope

The to-do list covering deliverables with agreed functionalities, quality, design etc



Schedule

The time frame for completion and delivery

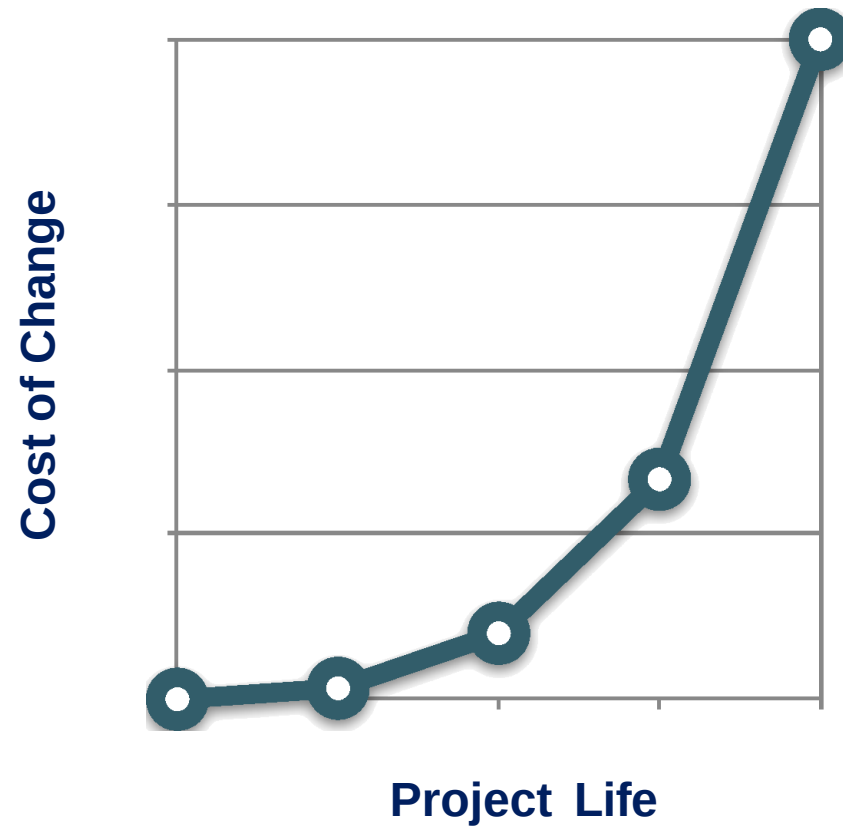
Budget

The financial frame for project execution

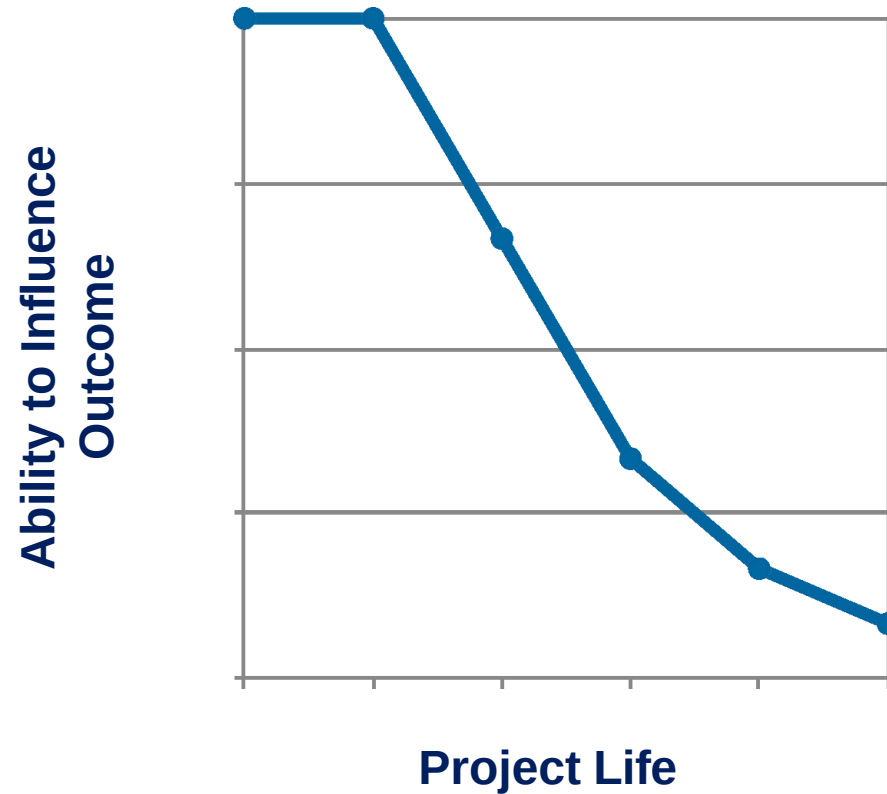
Scope Creep

- Scope creep refers to **uncontrolled harmful changes or growth in a project's scope** while the project is still in progress. This typically occurs when the scope of a project is not properly defined, documented, or controlled.
- Strategies: These are proper project plans, also referred to as **change management system** (*requests for changes become formal, and implications are more transparent*).

Cost of Change



Ability to Influence Project Outcome



Project Scope and Stakeholders

Project Scope and Stakeholders

A project scope must be discussed with, approved by and managed by the project main stakeholders

Project Stakeholders:

- Internal: Involved in executing the project
- External: Not involved in executing the project

Stakeholders Identification

Find a few questions below to help identify stakeholders:

- Who will/is funding the project?
- Who is working on the project?
- Who are the regulators in this context?
- Who will the project benefit?
- Who will be caused inconvenience during/after the project?
- Who are the suppliers of labour/material/equipment?
- Which media will report on the project etc

Project Scope and Stakeholders

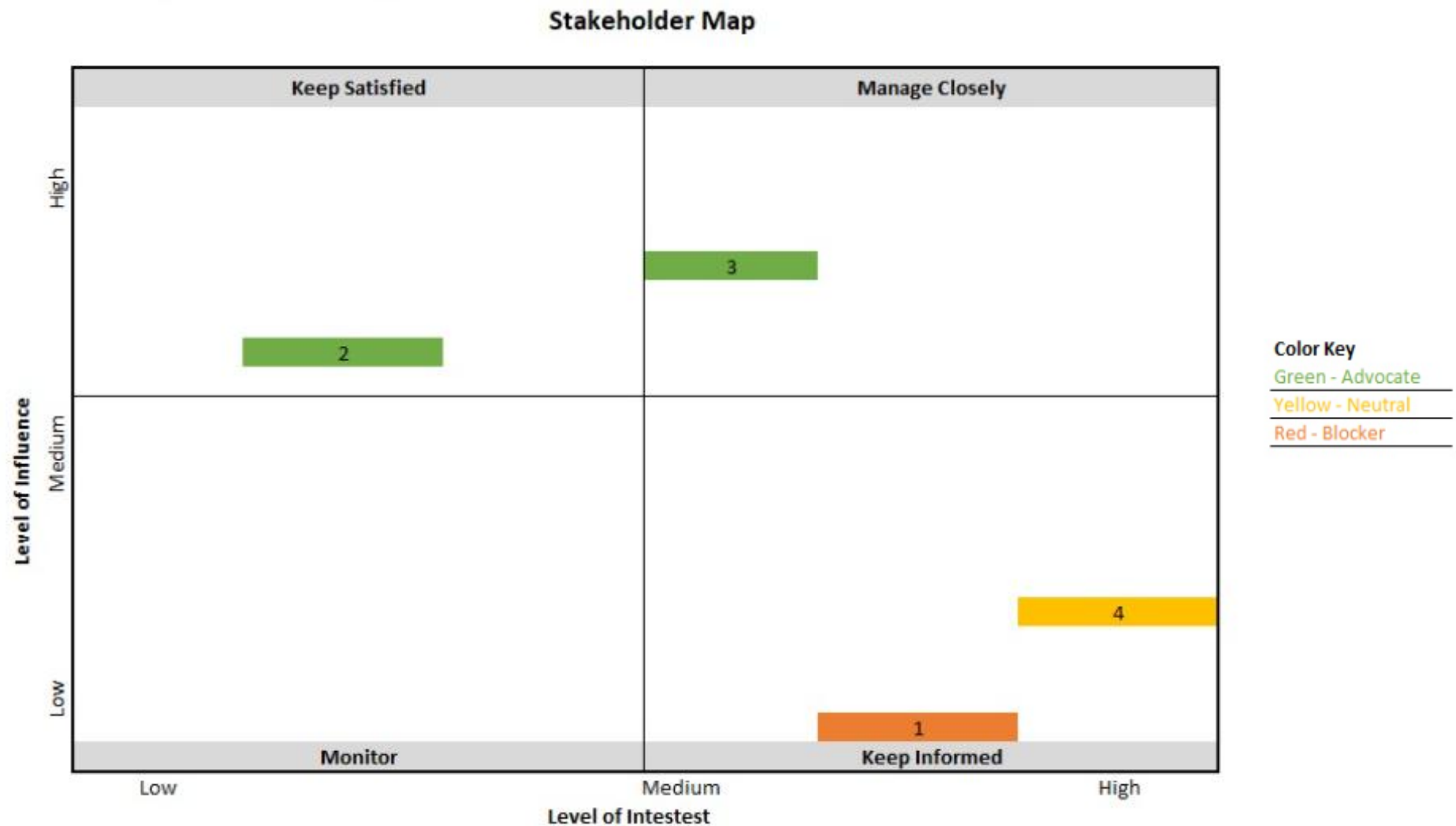
Some examples of Project Stakeholders:

- Project owners/initiators
- Project financiers/shareholders
- Project executives/managers
- Project team members (hired workers) etc

Stakeholder Relationship Management

- Often conflicting interests among stakeholders need to be resolved in the early stage of the project or large number of disturbing (and costly) changes can be expected at a later stage.
- Project Managers should always establish a stakeholder register and establish stakeholders strategies towards addressing issues that may arise sooner.
- Only after the stakeholders' analysis should the scope statement be discussed with and approved by the main stakeholders in a scoping workshop.

Stakeholders Map: Who Needs or Influences What?



<https://www.smartsheet.com/what-stakeholder-analysis-and-what-it-is-and-how-do-you-do-it-effectively>

Stakeholders Map: Who Needs or Influences What?



Business Model Canvas

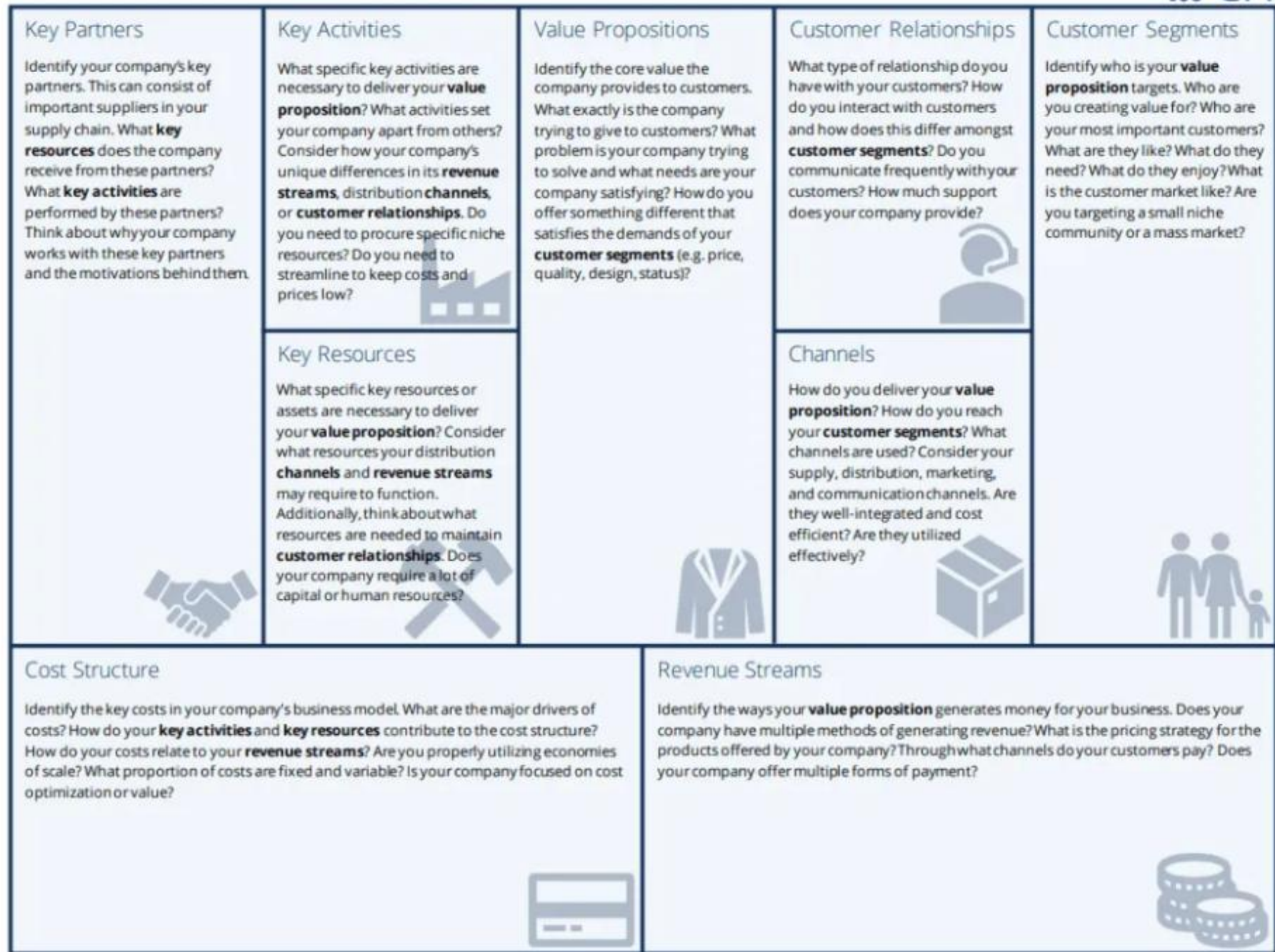
The Business Model Canvas (BMC) is a one-page strategic planning tool used by managers to illustrate, visualise, and analyse a business's key components, such as: value proposition, customer segmentation and relationships, revenue streams, key resources, activities, partners, and cost structure amongst others.

A business model canvas acts like an [executive summary](#) for the business plan.

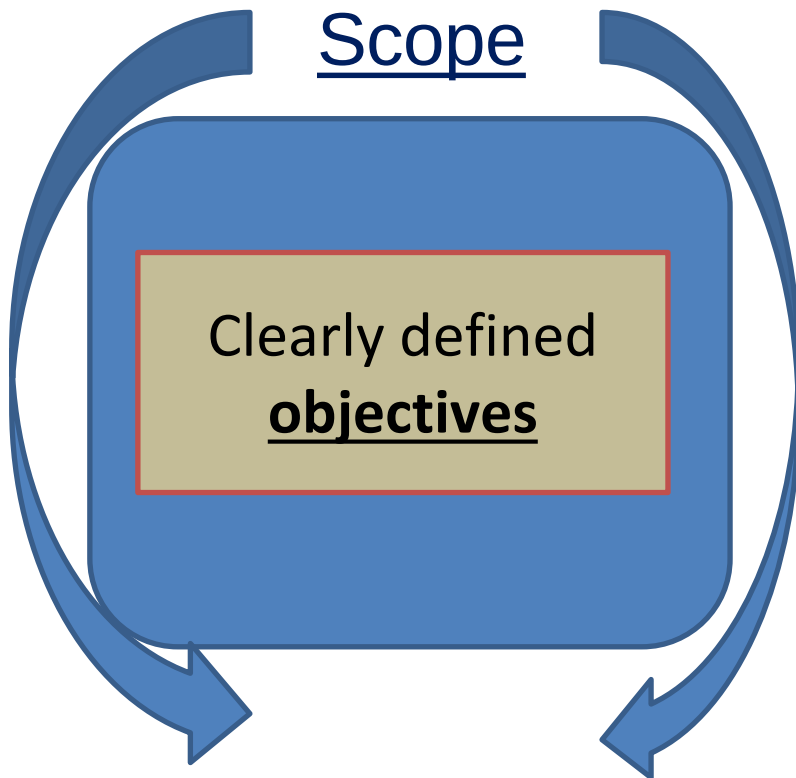
Core Components of a Business Model Canvas

- Key partners
- Key activities
- Value propositions
- Customer relationships
- Customer Segments
- Key Resources
- Channels
- Cost Structure
- Revenue Streams

Business Model Canvas



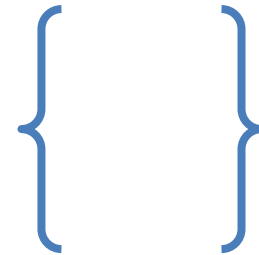
Business Model Canvas Critical Success Factors



limitations



delimitations



Learning Outcomes Evaluated

- Can you conveniently define the scope of your project, including creating an appealing scope statement?
- Can you intertwine the objectives→scope→limitations→ delimitations?
- From a personal point of view, which of the motivations related to poor project scoping resonates with you?
- Can you create a stakeholder map for your portfolio based on the requirements defined by the scope?
- Given a template, can you create a business canvas for your project?

For further reading, the underlisted are the resource materials:

Source: PMI (2017). A Guide to the Project Management Body of Knowledge (PMBOK Guide). 6th Edition.

Mirza, M. N., Pourzolfaghar, Z. and Shahnazari, M. (2013). Significance of Scope in Project Success. *Procedia Technology*. Vol. 9, pp. 722-729.

<https://www.smartsheet.com/what-stakeholder-analysis-and-mapping-and-how-do-you-do-it-effectively>

The End

Next lecture

- Unit 8: Project phases, deliverables and work breakdown structure



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